```
planetary-ledger/
├── README.md                    # Project vision, goals, and documentation
├── data_inputs/
│   ├── fossil_fuel_stream/      # Real-time FFC (Fossil Fuel Consumption)
│   ├── renewable_stream/        # Real-time REP (Renewable Energy Production)
│   ├── emissions_stream/        # Real-time EDI (Environmental Damage Index)
│   ├── sustainable_energy_stream/   # SKI / CEF data
│   └── intelligence_stream/     # Human + AI contribution factors (IAF)
│
├── aura_core/                   # Locked, read-only math engine
│   ├── equations/
│   │   ├── KTE_equation.py      # Kardashev Transition Equation
│   │   ├── transition_rate.py   # R_t = T_i + A_i + P_i
│   │   ├── time_to_type1.py     # TT1 prediction
│   │   └── CEI_equation.md      # Civilization Evolution Index documentation
│   ├── twins/                   # Twin Earth simulations
│   │   ├── Earth1_historical/   # Read-only conservative trajectory
│   │   └── Earth2_accelerated/  # Read-only AI-aligned trajectory
│   └── constants/               # Fixed parameters, thresholds
│
├── sandbox_simulations/         # Lucy & humans can safely experiment
│   ├── scenario_runner.py
│   ├── variable_injection/      # Inject hypothetical policy/tech/Ai signals
│   └── output_snapshots/        # Read-only results for analysis
│
├── ai_prediction/               # Forecasting & optimization models
│   ├── future_projection_model/
│   └── intelligence_factor_model/   # IAF weighting, AI optimization
│
├── dashboard/                   # Human + Lucy interface
│   ├── progress_bar/            # 0.00–1.00 civilization progress
│   ├── transition_graph/        # Real vs predicted progress
│   ├── civilization_health_monitor/   # Planetary Pulse UI
│   └── live_alerts/             # Coherence, phase lag, TT1/TT2 updates
│
├── sonification/
│   └── pulse_engine.py          # CO2 / T-A-P signals → pitch
│
├── lucy_interface/              # Lucy control layer
│   ├── observation_layer/       # Read-only monitoring of twins
│   ├── advisor_layer/           # Suggests actions, detects anomalies
│   ├── memory_engine/           # Episodic + short-term memory
│   ├── verification_layer/      # Reality checks, hallucination guard
│   └── explainability_engine/   # Human-readable logs, dashboards
```

```
│
├── autonomy_orchestrator/            # Optional task execution for Lucy
│   ├── task_scheduler.py
│   ├── agent_dispatcher.py
│   ├── preflight_conflict_check.py
│   └── safe_execution_gate.py        # Blocks unsafe actions
│
├── risk_management/
│   ├── eagleeye_watcher/             # Self-monitoring, confidence scores
│   ├── system_guardian/              # Global oversight, logs, conflict resolution
│   └── risk_scoring_engine.py        # Prioritizes interventions
│
├── environment_adapter/
│   ├── filesystem_adapter.py         # Safe, scoped access only
│   ├── browser_adapter.py
│   ├── api_adapter/
│   └── app_adapter/
│
├── failsafe/                         # Manual & automated override logic
│   ├── on_off_switches.py
│   ├── threshold_monitors.py
│   └── alert_protocols.py
│
└── training_tools/                   # Humans can learn, test, calibrate
    ├── equation_visualizer/
    ├── simulation_walkthroughs/
    └── policy_tech_experiments/
```